

When to compute the utility: a bound on the pair (μ_g, σ_g)

At a query image the GP gives a Gaussian posterior on the log firing rate, $g \sim \mathcal{N}(\mu_g, \sigma_g^2)$, and the spike count in the window is Poisson with rate e^g (same setup as `laplace_explainer.tex`). We want a rule that decides, *before* the utility is computed, whether the posterior at an image is sane enough for that utility to mean anything. The rule is a bound on the *pair* (μ_g, σ_g) , not on either one alone.

1 A small σ_g is not a small spread

σ_g is uncertainty in the *log* rate, so it acts on the rate multiplicatively: one step of σ_g multiplies or divides the rate by e^{σ_g} . A pair (μ_g, σ_g) that looks modest on paper can hide a very skewed, very wide distribution of actual spike counts.

Take $\mu_g = 3$, $\sigma_g = 1.5$. The most probable count is just 2 spikes and the median is 20 — both look like an ordinary cell — yet the mean is 62, the standard deviation 180, and the $+3\sigma$ rate edge $e^{\mu_g+3\sigma_g} = e^{7.5} \approx 1800$ spikes in the ~ 320 ms window (≈ 5600 Hz, impossible). The most likely count and the mean differ thirty-fold.

The three location measures of the count distribution $p(r)$ track those of the underlying log-normal rate,

$$\text{mode} \approx e^{\mu_g - \sigma_g^2}, \quad \text{median} \approx e^{\mu_g}, \quad \text{mean} = e^{\mu_g + \sigma_g^2/2},$$

and they fan apart as σ_g grows: the median stays pinned at the typical rate e^{μ_g} , while the mode slides down and the mean and the upper tail run away.

μ_g	σ_g	σ_g^2	mode	median	mean	std	$e^{\mu_g+3\sigma_g}$
1	1.0	1.00	1	3	4	6	55
2	1.0	1.00	3	7	12	16	148
3	1.0	1.00	7	20	33	44	403
4	1.0	1.00	20	55	90	118	1097
1	1.5	2.25	0	3	8	25	245
2	1.5	2.25	1	7	23	66	665
3	1.5	2.25	2	20	62	180	1808
4	1.5	2.25	6	55	168	490	4915

mode and median computed from $p(r)$ by deterministic quadrature; mean and std are closed form; $e^{\mu_g+3\sigma_g}$ is the gate quantity of §2. Read across a row: at fixed μ_g , raising σ_g from 1 to 1.5 barely moves the median but multiplies the standard deviation and the $+3\sigma$ count several-fold. A pair can look reasonable by its typical count and still encode a distribution whose mean and upper tail are physically impossible.

2 The gate

The quantity that has to stay physical is the high end of the rate the posterior still finds plausible, the $+3\sigma$ edge $e^{\mu_g+3\sigma_g}$. A retinal ganglion cell produces at most of order a hundred spikes in the counting window, so we require

$$\boxed{e^{\mu_g+3\sigma_g} < T} \quad \Longleftrightarrow \quad \mu_g + 3\sigma_g < \ln T.$$

This couples the two parameters: a high mean leaves less room for the spread, and vice versa. In the (σ_g^2, μ_g) plane the boundary $\mu_g = \ln T - 3\sqrt{\sigma_g^2}$ is a curve; in the (σ_g, μ_g) plane it is the straight

line $\mu_g = \ln T - 3\sigma_g$. The multiplier 3 is a margin choice: it asks the 99.9th rate percentile to stay below T ; using 2 would test only the ~ 98 th, and Poisson noise pushes the true count a little past the rate edge in any case.

3 What T should be: tie it to $r_{\max}$

T is not free. The utility truncates the Poisson sum at $r_{\max}$ (it sums over $r = 0, \dots, r_{\max}$), so the computed entropies are correct only if the count distribution has negligible mass above $r_{\max}$. The edge $e^{\mu_g + 3\sigma_g}$ estimates where that mass ends, so the self-consistent gate is

$$e^{\mu_g + 3\sigma_g} < r_{\max}.$$

The live experiment used $r_{\max} = 100$. With that truncation the honest threshold is $T \approx 100$: above it the sum cannot see the counts the posterior places there, and the utility is computed but wrong. Setting $T = 1000$ is consistent only if $r_{\max}$ is also raised to ≈ 1000 . The clean choice is to fix $r_{\max}$ from physics — a few hundred counts, a few hundred Hz — and set $T = r_{\max}$. The gate and the truncation move together.

4 A second reason, independent of the counts

Even where the counts are physical, a large σ_g degrades the Laplace approximation the utility is built on (`laplace_explainer.tex`): a small σ_g pins the integrand peak and the approximation is near-exact, a large σ_g widens it and the error grows. So past the gate the utility is suspect twice over — the distribution is unphysical *and* the number computed from it is a poor approximation.

A flat cap $\sigma_g \leq 1.5$ is the single-number shortcut for this gate; it is conservative at low μ_g and too loose near $\mu_g \approx 4$, because the real condition is on the pair $\mu_g + 3\sigma_g$.